

输入电路

$$\frac{V_i - V_F}{R} + \frac{0 - V_F}{R} = \frac{V_F - (\frac{V_{o1}}{2} + V_B)}{R}$$

$$\frac{V_S - V_T}{R} + \frac{0 - V_T}{R} = \frac{V_T - V_B}{R}$$

$$V_F = V_T$$

$$2(V_S - V_i) = V_{o1}$$

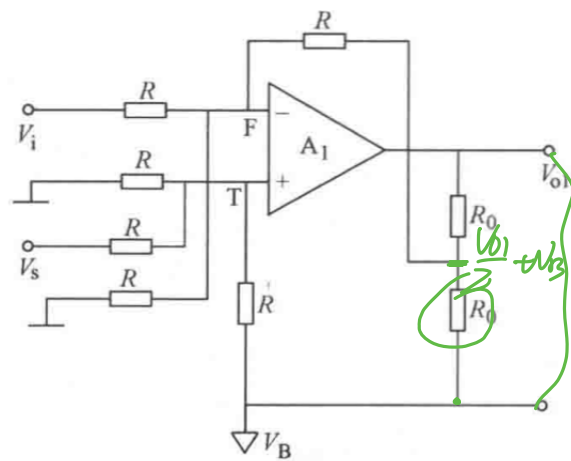
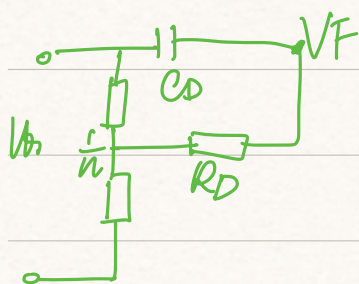


图 3-9 输入电路图

比例微分电路

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$$V_T = I_{D(s)} R_D + \frac{1}{n} V_{o1}$$

$$I_{D(s)} = (1 - \frac{1}{n}) V_{o1} / (\frac{1}{C_D s} + R_D)$$

$$V_T = \frac{(n R_D C_D s + 1) V_{o1}}{n (1 + R_D C_D s)}$$

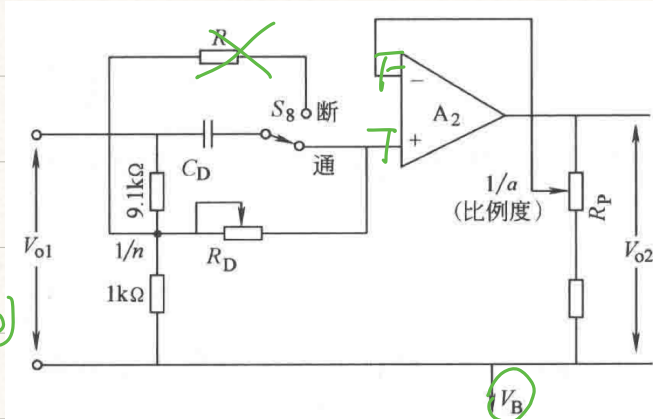
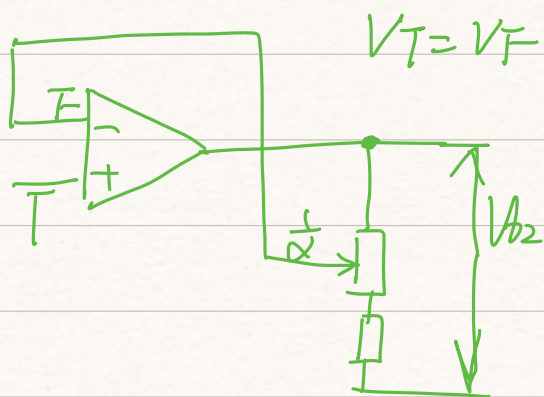


图 3-10 比例微分电路



$$V_T = V_F$$

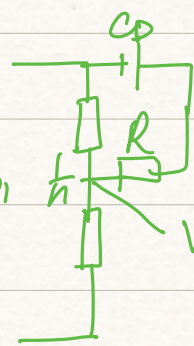
$$\frac{V_{o2}}{\alpha} = V_F$$

$$V_{o2} = \frac{\alpha}{n} \frac{(n R_D C_D s + 1)}{(1 + R_D C_D s)} V_{o1}$$

$$n = K_D \quad n R_D C_D = T_D$$

$$V_{o2}(s) = \frac{\alpha}{n} \frac{T_D s + 1}{1 + \frac{T_D}{K_D} s} V_{o1}(s)$$

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$$V_T = \frac{1}{n} V_{o1} = V_F$$

$$V_F = \frac{1}{\alpha} V_{o2}$$

$$V_{o2} = \frac{\alpha}{n} V_{o1}$$

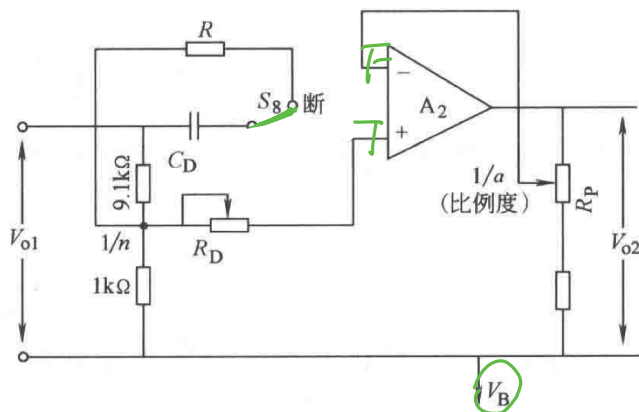


图 3-10 比例微分电路

比例积分电路

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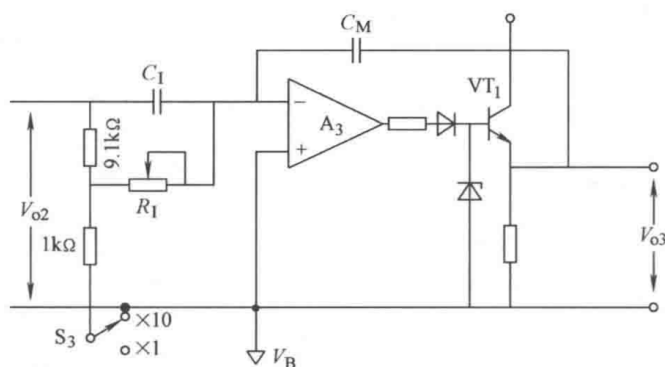
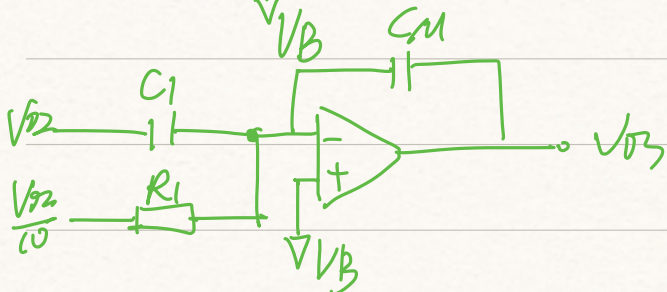
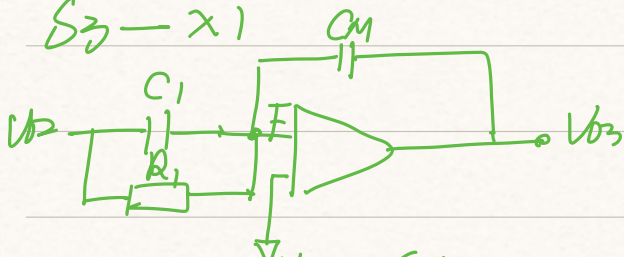


图 3-12 比例积分电路

$$\begin{cases} \frac{V_{o2} - V_F}{\frac{1}{C_1 s}} + \frac{V_{o2/m} - V_F}{R_1} = \frac{V_F - V_{o3}}{\frac{1}{C_M s}} \\ V_{o3} = -K V_F \end{cases}$$

\$\Rightarrow\$

$$C_1 s (V_{o2} + \frac{V_{o3}}{K}) + \frac{\frac{V_{o2}}{m} + \frac{V_{o3}}{K}}{R_1} + C_M s (\frac{V_{o3}}{K} + V_{o3}) \Rightarrow$$

$$(C_1 s + \frac{1}{m R_1}) V_{o2} + (\frac{C_1 s}{K} + \frac{1}{K R_1} + \frac{C_M s}{K} + C_M s) V_{o3} \Rightarrow$$

$$1 + \frac{1}{m R_1 C_1 s}$$

$$1 + \frac{1}{K s}$$

$$\frac{1}{K} + \frac{1}{KR_1G_S} + \frac{C_M}{C_1} + \frac{C_M}{C_1}$$

$$1 + \frac{1}{T_2 R_2 S}$$

$$= - \frac{C_M}{C_1} \left(\frac{C_1}{K C_M} + \frac{1}{KR_1 G_S} + \frac{1}{K} + 1 \right)$$

$$= - \frac{C_1}{C_M} \frac{1 + \frac{1}{m R_1 C_1 S}}{1 + \frac{1}{K} \left(\frac{C_1}{C_M} + 1 \right) + \frac{1}{KR_1 G_S}}$$

$$\text{因 } K \gg 10^5 \text{ 故 } \frac{1}{K} \left(\frac{C_1}{C_M} + 1 \right) \ll 1$$

$$= - \frac{C_1}{C_M} \frac{1 + \frac{1}{m R_1 C_1 S}}{1 + \frac{1}{KR_1 G_S}}$$

$$\text{令 } T_2 = m R_1 C_1 \quad K_2 = \frac{K C_M}{m C_1}$$

$$\frac{V_{02}(S)}{V_{01}(S)} = - \frac{C_1}{C_M} \frac{1 + \frac{1}{T_2 S}}{1 + \frac{1}{T_2 K_2 S}}$$

例 4-1 某单容液位过程如图 4-7 所示。该过程中，储罐中的液位高度 h 为被控参数（即过程的输出），流入储罐的体积流量 q_1 为过程的输入量， q_1 的大小可通过阀门 1 的开度来改变；流出储罐的体积流量 q_2 为中间变量（即为过程的干扰），它取决于用户需要，其大小可以通过阀门 2 的开度来改变。试确定 h 与 q_1 之间的数学关系。

$$\textcircled{1} q_1 - q_2 = A \frac{dh}{dt}$$

$$\Delta q_1 - \Delta q_2 = A \frac{d\Delta h}{dt} \quad , A \text{ 为储罐底面积}$$

$$\Delta q_2 = \frac{\Delta h}{R_2}$$

$$q_{10} = q_{20} \text{ 时, 有 } \Delta h = 0$$

$$\Delta q_1 - \Delta h / R_2 = A \frac{d\Delta h}{dt}$$

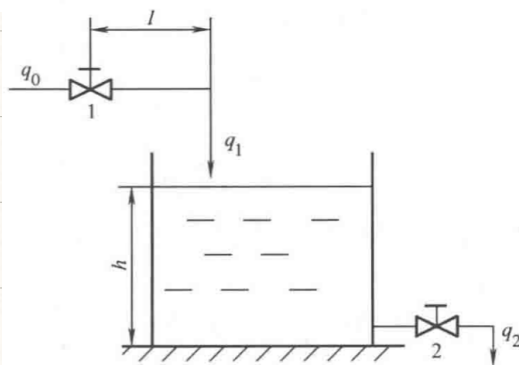


图 4-7 单容液位过程

$$Q_1(s) = \left(\frac{1}{R_2} + As \right) H(s)$$

$$\frac{H(s)}{Q_1(s)} = \frac{R_2}{1 + R_2 C s} = \frac{K}{1 + T s} \quad K = R_2, T = R_2 A$$

$$\begin{aligned} \textcircled{2} \quad q_0(t-\tau) - q_2 &= A \frac{dh}{dt} \\ \Delta q_0(t-\tau) - \Delta q_2 &= A \frac{d\Delta h}{dt} \\ \Delta q_2 &= \frac{\Delta h}{R_2} \\ \Delta q_0(t-\tau) &= \frac{\Delta h}{R_2} + A \frac{d\Delta h}{dt} \\ Q_0(s) e^{-Ts} &= \left(\frac{1}{R_2} + As \right) H(s) \end{aligned}$$

$$\frac{H(s)}{Q_0(s)} = \frac{R_2}{1 + R_2 C s} e^{-Ts}$$

例 4-2 某单容热力过程如图 4-9 所示。加热装置采用电能加热，给容器输入热流量 q_i 。容器的热容为 C ，容器中液体的比热容为 c_p 。流量为 q 的液体以 T_i 的入口温度流入，以 T_c 的出口温度流出（ T_c 同时也是容器中液体的温度）。设容器所在的环境温度为 T_0 。试求该过程的输出量 T_c 与热流量 q_i 、液体入口温度 T_i 以及环境温度 T_0 之间的数学关系。

$$q c_p T_i - q c_p T_c + q_i - (T_c - T_0) A k_r = C \frac{dT_c}{dt}$$

$$q c_p \Delta T_i - q c_p \Delta T_c + \Delta q_i - (T_c - T_0) A k_r = C \frac{d\Delta T_c}{dt}$$

A — 容器换热表面积, k_r — 保温材料的热导系数

$$\begin{aligned} q c_p T_i(s) - q c_p T_c(s) + Q_i(s) - A k_r T_c(s) + A k_r T_0(s) \\ = C s T_c(s) \end{aligned}$$

$$T_c(s) = \frac{q c_p}{C s + A k_r + q c_p} T_i(s) + \frac{Q_i(s)}{C s + A k_r + q c_p} + \frac{A k_r}{C s + A k_r + q c_p} T_0(s)$$

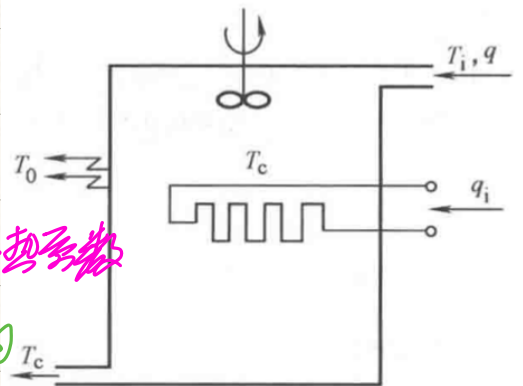


图 4-9 单容热力过程

$$q c_p = K_p \quad A k_r = \frac{1}{k}$$

$$T_c(s) = \frac{kp}{Cs + kp + \frac{1}{R}} T_i(s) + \frac{Q_{i1}(s)}{Cs + kp + \frac{1}{R}} + \frac{\frac{R}{C}}{Cs + kp + \frac{1}{R}} T_o(s)$$

$$= \frac{\frac{kpR}{1+Rkp}}{\frac{CRs}{1+Rkp} + 1} T_i(s) + \frac{\frac{R}{1+Rkp}}{\frac{CRs}{1+Rkp} + 1} Q_{i1}(s) + \frac{\frac{1}{1+Rkp}}{\frac{CRs}{1+Rkp} + 1} T_o(s)$$

例 4-3 在图 4-7 中, 如果将阀 2 换成定量泵, 使输出流量 q_2 在任何情况下都与液位 h 的大小无关, 如图 4-10 所示。试求 h 与 q_1 之间的关系。

$$q_1 - q_2 = C \frac{dh}{dt}$$

$$\Delta q_1 - \Delta q_2 = C \frac{d\Delta h}{dt}$$

$$\Delta q_2 = 0$$

$$Q_1(s) = CS H(s)$$

$$\frac{H(s)}{Q_1(s)} = \frac{1}{CS} = \frac{1}{TS}, T = C$$

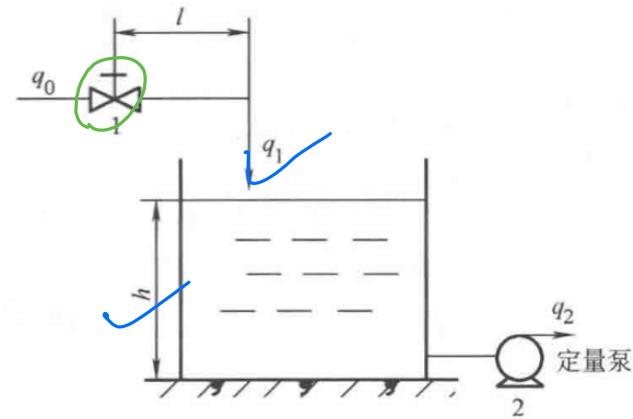


图 4-10 阀 2 改为定量泵的液位过程

例 4-4 图 4-12 所示为一分离式双容液位过程。图 4-12a 中, 设 q_1 为过程输入量, 第二个液位槽的液位 h_2 为过程输出量, 若不计第一个与第二个液位槽之间液体输送管道所形成的时间延迟, 试求 h_2 与 q_1 之间的数学关系。

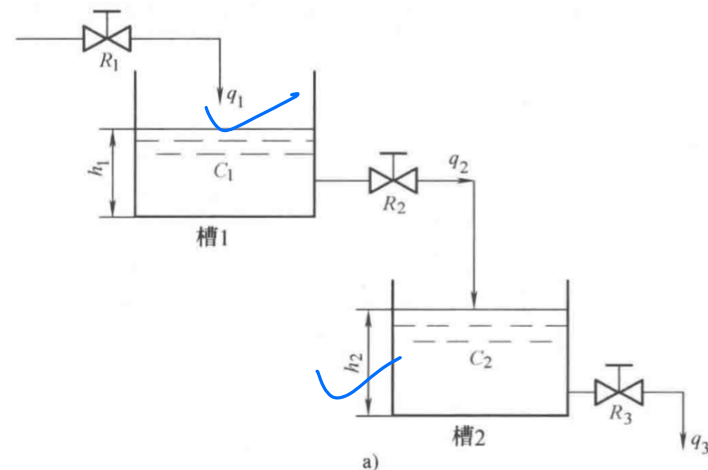
$$q_1 - q_2 = C_1 \frac{dh_1}{dt}$$

$$\Delta q_1 - \Delta q_2 = C_1 \frac{d\Delta h_1}{dt}$$

$$\Delta q_2 = \frac{\Delta h_1}{R_2}$$

$$Q_1(s) - Q_2(s) = C_1 R_2 S Q_2(s)$$

$$\frac{Q_2(s)}{Q_1(s)} = \frac{1}{1 + C_1 R_2 S}$$



$$q_2 - q_3 = C_2 \frac{dh_2}{dt}$$

$$\Delta q_2 - \Delta q_3 = C_2 \frac{d\Delta h_2}{dt}$$

$$\Delta q_3 = \frac{\Delta h_2}{R_3}$$

$$h_2(s) - \frac{h_2(s)}{R_3} = C_2 s h_2(s)$$

$$\frac{h_2(s)}{h_2(s)} = \frac{R_3}{R_3 C_2 s + 1}$$

$$\frac{H_2(s)}{H_1(s)} = \frac{1}{C_1 R_2 s + 1} \frac{R_3}{C_2 R_3 s + 1} = \frac{k}{(T_1 s + 1)(T_2 s + 1)}, \quad k = R_3, \quad T_1 = C_1 R_2, \quad T_2 = C_2 R_3$$

若 q_3 为定量泵

$$q_1 - q_2 = C_1 \frac{dh_1}{dt}$$

$$\Delta q_1 - \Delta q_2 = C_1 \frac{d\Delta h_1}{dt}$$

$$\Delta q_2 = \frac{\Delta h_1}{R_2}$$

$$h_1(s) - h_2(s) = C_1 R_2 s h_2(s)$$

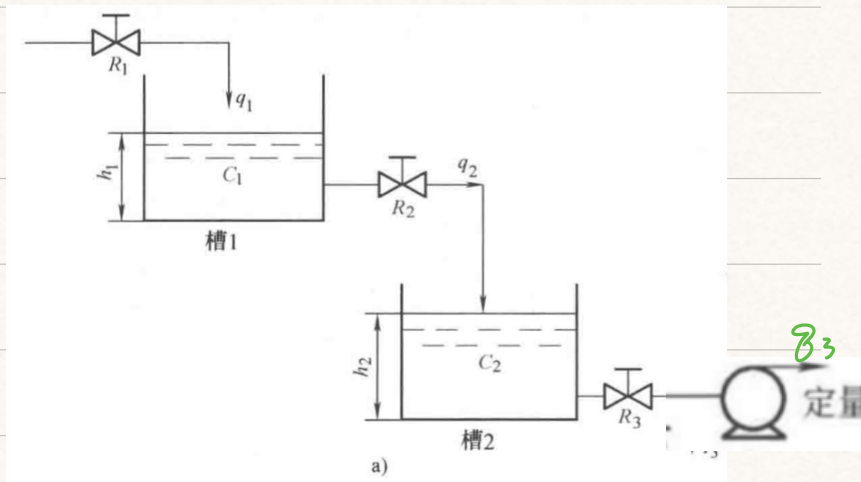
$$\frac{h_2(s)}{h_1(s)} = \frac{1}{C_1 R_2 s + 1}$$

$$\Delta q_2 - \Delta q_3 = C_2 \frac{d\Delta h_2}{dt}$$

$$\Delta q_3 = 0$$

$$h_2(s) = C_2 s h_2(s)$$

$$\frac{h_2(s)}{h_2(s)} = \frac{1}{C_2 s}$$



$$\frac{H(s)}{Q_1(s)} = \frac{1}{C_1 R_2 s + 1} \frac{1}{C_2 s} = \frac{1}{T_1 s (T_2 s + 1)}, \quad T_1 = C_2, \quad T_2 = C_1 R_2$$

例 4-5 图 4-13 为一个并联式双容液位过程。与图 4-12 相比， q_2 的大小不仅与槽 1 的液位 h_1 有关，而且与槽 2 的液位 h_2 也有关。设图中各个变量及参数的意义与例 4-4 相同，试求 h_2 与 q_1 之间的数学关系。

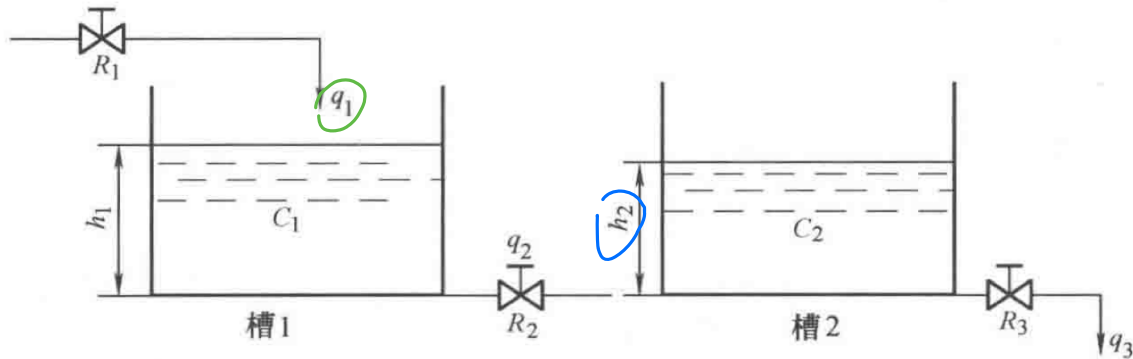


图 4-13 并联式双容液位过程

$$q_1 - q_2 = C_1 \frac{dh_1}{dt}$$

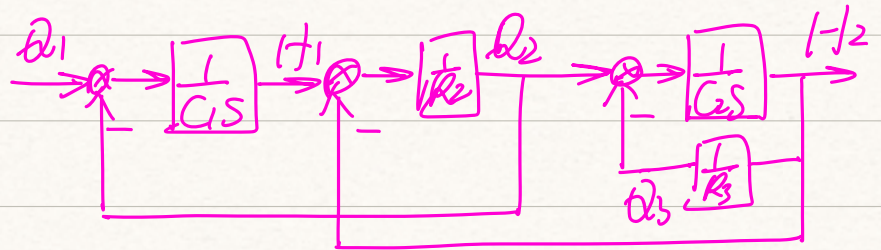
$$\Delta q_1 - \Delta q_2 = C_1 \frac{d\Delta h_1}{dt}$$

$$\Delta q_2 = \frac{\Delta h_1 - \Delta h_2}{R_2}$$

$$\Delta q_2 - \Delta q_3 = C_2 \frac{d\Delta h_2}{dt}$$

$$\Delta q_3 = \frac{\Delta h_2}{R_3}$$

$$\frac{H(s)}{Q_1(s)} = \frac{R_3}{C_1 C_2 R_2 R_3 s^2 + [C_1 R_2 + (C_1 + C_2) R_3] s + 1}$$



(1) 如图 4-32 所示， q_1 为过程的流入量， q_2 为流出量， h 为液位高度， C 为容量系数。若以 q_1 为过程的输入量， h 为输出量（被控量），设 R_1 、 R_2 为线性液阻，求过程的传递函数 $G_0(s) = H(s)/Q_1(s)$ 。

$$q_1 - q_2 = C \frac{dh}{dt}$$

$$\Delta q_1 - \Delta q_2 = C \frac{d\Delta h}{dt}$$

$$\Delta q_2 = \frac{\Delta h}{R_2}$$

$$Q_1(s) - \frac{1}{R_2} H(s) = CS H(s)$$

$$\frac{H(s)}{Q_1(s)} = \frac{R_2}{CR_2s + 1}$$

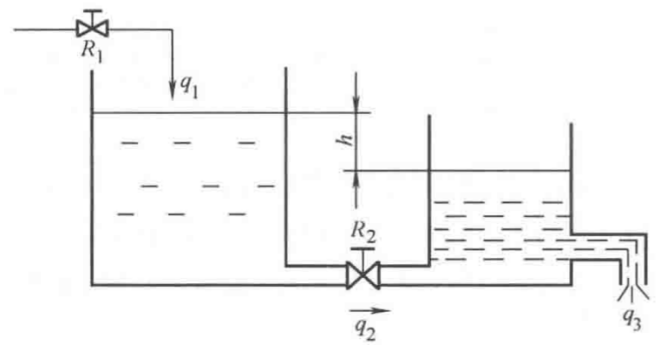
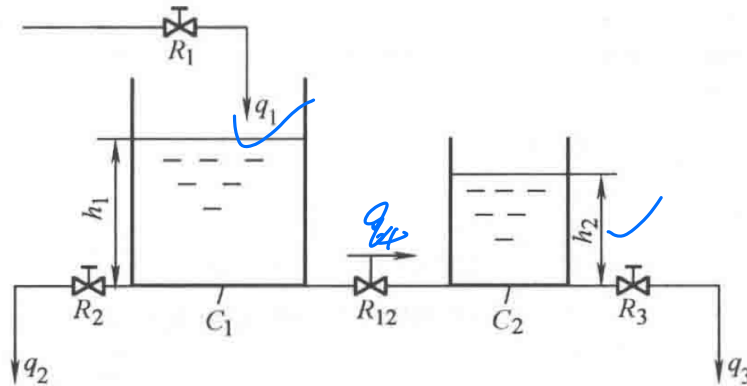


图 4-32 综合练习题 (1) 液位过程

(2) 已知两个水箱串联工作 (如图 4-33 所示), 其输入量为 q_1 , 流出量为 q_2, q_3 , h_1, h_2 分别为两个水箱的水位, h_2 为被控参数, C_1, C_2 为其容量系数, 假设 R_1, R_2, R_{12}, R_3 为线性液阻。要求:

- 1) 列写该液位过程的微分方程组。
- 2) 画出该过程的框图。
- 3) 求该液位过程的传递函数 $G_0(s) = H_2(s)/Q_1(s)$ 。



$$q_1 - q_2 - q_4 = C_1 \frac{dh_1}{dt} \quad \checkmark$$

$$\Delta q_1 - \Delta q_2 - \Delta q_4 = C_1 \frac{d\Delta h_1}{dt} \quad \checkmark$$

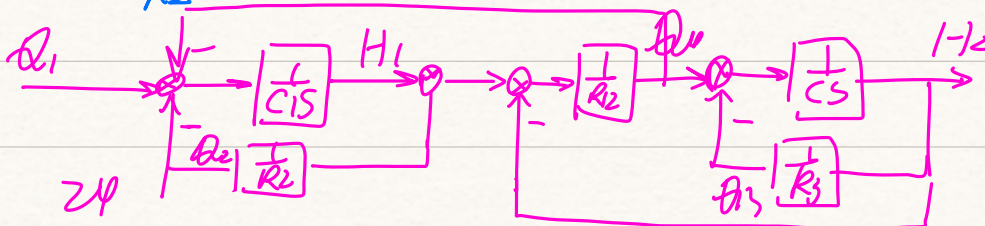
$$\Delta q_4 = \frac{\Delta h_1 - \Delta h_2}{R_{12}} \quad \checkmark$$

$$\Delta q_2 = \frac{\Delta h_1}{R_2} \quad \checkmark$$

$$q_4 - q_3 = C_2 \frac{dh_2}{dt} \quad \checkmark$$

$$\Delta q_4 - \Delta q_3 = C_2 \frac{d\Delta h_2}{dt} \quad \checkmark$$

$$\Delta q_3 = \frac{\Delta h_2}{R_3} \quad \checkmark$$



(3)

$$\Delta g_1 - \Delta g_2 - \Delta g_4 = C_1 \frac{d\Delta h_1}{ds}$$

$$\Delta g_4 = \frac{\Delta h_1 - \Delta h_2}{R_{12}}$$

$$\Delta h_1 = R_{12} \Delta g_4 + \Delta h_2$$

$$\Delta g_2 = \frac{\Delta h_1}{R_2}$$

$$\Delta g_1 - \frac{\Delta h_1}{R_2} - \Delta g_4 = C_1 \frac{d(R_{12} \Delta g_4 + \Delta h_2)}{ds}$$

$$\Delta g_1 - \Delta g_4 - \frac{1}{R_2} (R_{12} \Delta g_4 + \Delta h_2) = C_1 R_{12} S \Delta g_4 + C_1 S \Delta h_2$$

$$\Delta g_1 = \left(1 + \frac{R_{12}}{R_2} + C_1 R_{12} S\right) \frac{R_3 C_2 S + 1}{R_3} H_{2(s)} + \left(C_1 S + \frac{1}{R_2}\right) H_2$$

$$\Delta g_1 = \frac{1}{R_3} \left[R_3 C_2 S + 1 + \frac{R_{12}}{R_2} (R_3 C_2 S + 1) + C_1 R_{12} S (R_3 C_2 S + 1) + \left(C_1 S + \frac{R_3}{R_2}\right) H_2 \right]$$

$$= \frac{1}{R_3} \left[R_3 C_2 S + 1 + \frac{R_{12} R_3 C_2}{R_2} S + \frac{R_{12}}{R_2} + R_{12} R_3 C_1 C_2 S^2 + R_{12} C_1 S + \frac{R_3}{R_2} C_1 S + \frac{R_3}{R_2} \right] H_2$$

$$= \frac{1}{R_3} \left(R_{12} R_3 C_1 C_2 S^2 + \left(R_3 C_2 + \frac{R_{12} R_3 C_2}{R_2} + R_{12} C_1 + C_1 R_3 \right) S + \left(\frac{R_3}{R_2} + \frac{R_{12}}{R_2} + 1 \right) \right) H_2$$

$$\frac{H_2(s)}{\Delta g_1(s)} = \frac{R_2 R_3}{R_{12} R_2 R_3 C_1 C_2 S^2 + [(R_2 + R_{12}) R_3 C_2 + (R_{12} + R_3) C_1] S + (R_3 + R_{12} + R_2)}$$

例 7-1 已知某比值控制系统，采用孔板和差压变送器测量主、副流量，主流量变送器的最大量程为 $q_{1\max} = 12.5 \text{ m}^3/\text{h}$ ，副流量变送器的最大量程为 $q_{2\max} = 20 \text{ m}^3/\text{h}$ ，生产工艺要求 $q_2/q_1 = K = 1.4$ ，试计算：

1) 不加开方器时，DDZ-III型仪表的比值系数 K' 。

2) 加开方器后，DDZ-III型仪表的比值系数 K' 。

$$1) K' = \left(K \frac{q_{1\max}}{q_{2\max}} \right)^2 = 0.766$$

$$2) K' = K \frac{q_{1\max}}{q_{2\max}} = 0.875$$

输入电路

$$\begin{cases} \frac{V_i - V_F}{R} + \frac{0 - V_F}{R} = \frac{V_F - (V_B + \frac{V_{o1}}{2})}{R} \\ \frac{V_s - V_T}{R} + \frac{0 - V_T}{R} = \frac{V_T - V_B}{R} \\ V_T = V_F \\ V_{o1} = 2(V_s - V_i) \end{cases}$$

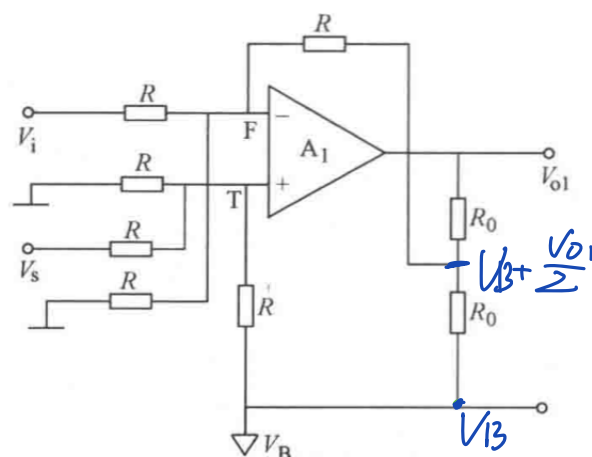


图 3-9 输入电路图

比例微分电路

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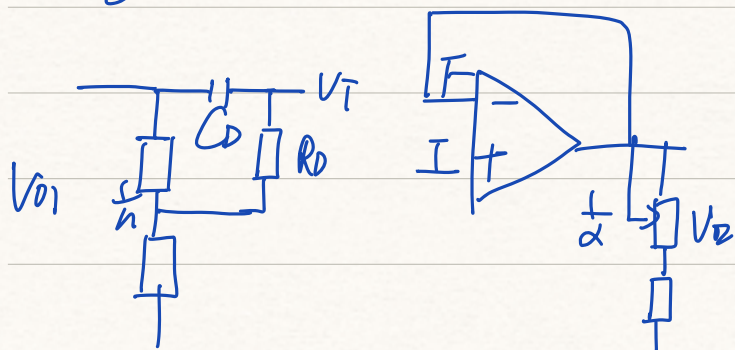


图 3-10 比例微分电路

$$V_T = \frac{V_{o1}}{n} + \frac{R_D}{R_D + \frac{1}{C_D s}} \left(1 - \frac{1}{n}\right) V_{o1}$$

$$V_T = V_F = \frac{1}{\alpha} V_{o2}$$

$$\frac{1}{\alpha} V_{o2} = \frac{V_{o1}}{n} + \frac{R_D C_D s}{R_D C_D s + 1} \frac{n-1}{n} V_{o1}$$

$$= \frac{1 + n R_D C_D s}{n (R_D C_D s + 1)} V_{o1}$$

$$\frac{V_{o2}}{V_{o1}} = \frac{\alpha}{n} \frac{1 + n R_D C_D s}{1 + R_D C_D s} \quad \begin{matrix} T_D = n R_D C_D \\ K_D = \frac{1}{n} \end{matrix} \quad \frac{\alpha}{n} \frac{1 + T_D s}{1 + \frac{T_D s}{K_D}}$$

S8 断

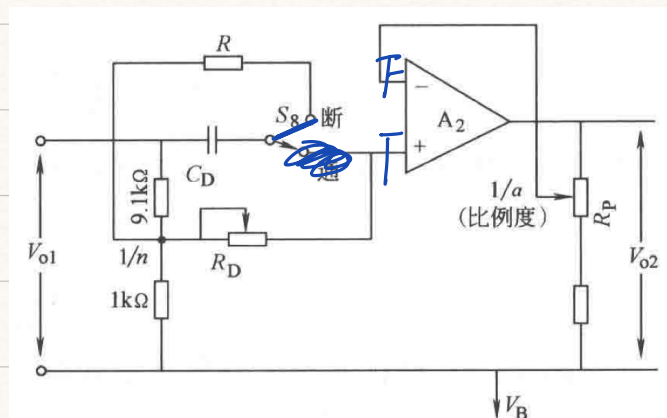
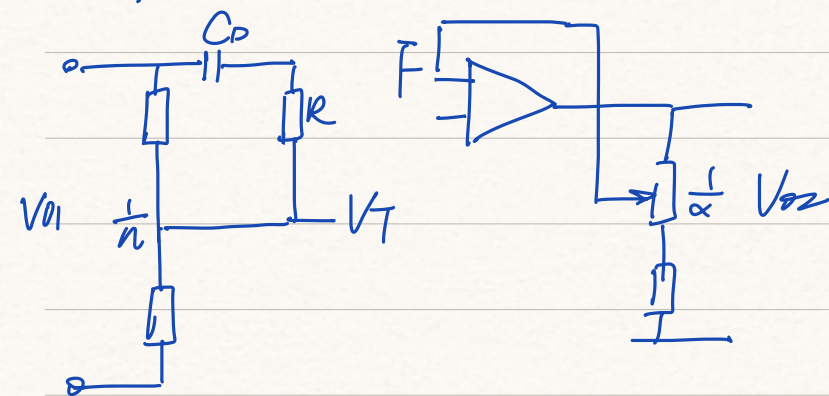


图 3-10 比例微分电路

$$V_T = \frac{1}{n} V_{o1}$$

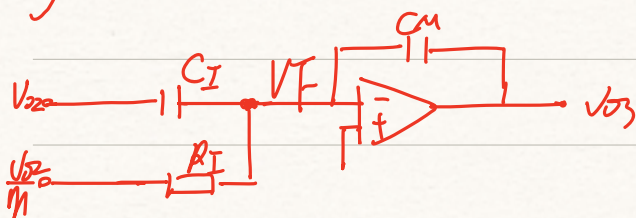
$$V_T = V_F$$

$$V_F = \frac{1}{\alpha} V_{o2}$$

$$\frac{1}{n} V_{o1} = \frac{1}{\alpha} V_{o2}$$

$$\frac{V_{o2}}{V_{o1}} = \frac{\alpha}{n}$$

比例积分电路



$$\frac{V_{i2} - V_F}{C_I s} + \frac{V_{i2} - V_F}{R_I} = \frac{V_F - V_{o3}}{\frac{1}{C_M s}}$$

$$V_{o3} = -K V_F$$

$$C_I s \left(V_{i2} + \frac{V_{o3}}{K} \right) + \frac{1}{R_I} \left(\frac{V_{i2}}{m} + \frac{V_{o3}}{K} \right) = C_M s \left(-\frac{V_{o3}}{K} - V_{o3} \right)$$

$$\left(C_I s + \frac{1}{R_I m} \right) V_{i2} = \left[-C_M s \left(\frac{1}{K} + 1 \right) - \frac{1}{R_I K} - \frac{C_I s}{K} \right] V_{o3}$$

$$\frac{V_{o3}}{V_{i2}} = - \frac{C_I s + \frac{1}{R_I m}}{C_M \left(\frac{1}{K} + 1 \right) s + \frac{1}{R_I K} + \frac{C_I s}{K}}$$

$$= - \frac{1 + \frac{1}{R_I m C_I s}}{\left(\frac{1}{K} + 1 \right) + \frac{1}{C_M R_I K s} + \frac{C_I}{K C_M}} \frac{C_I}{C_M}$$

$$= - \frac{C_I}{C_M} \frac{1 + \frac{1}{R_I m C_I s}}{1 + \frac{1}{C_M R_I K s} + \frac{1}{K} \left(1 + \frac{C_I}{C_M} \right)}$$

$$K \gg 10^5 \quad \frac{1}{K} \left(1 + \frac{C_I}{C_M} \right) \ll 1$$

$$\text{原式} = - \frac{C_I}{C_M} \frac{1 + \frac{1}{R_I m C_I s}}{1 + \frac{1}{C_M R_I K s}}$$

$$K_C = - \frac{C_I}{C_M} \quad T_I = m R_I C_I \quad K_I = \frac{K C_M}{m C_I}$$

$$= K_C \frac{1 + \frac{1}{T_I s}}{1 + \frac{1}{T_I K_C s}}$$

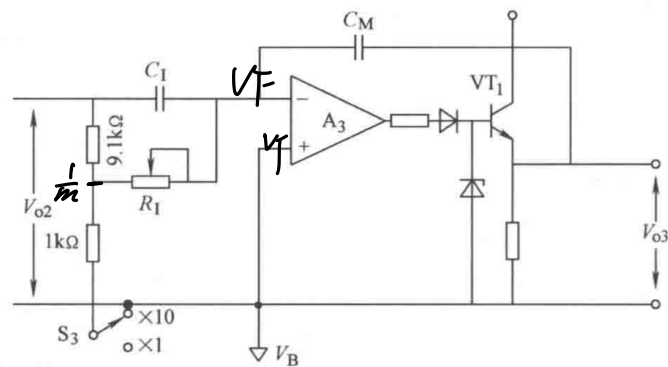


图 3-12 比例积分电路